



- 1 (a) Define acceleration.

.....  
 ..... [1]

- (b) A small aircraft is flying horizontally at a speed of  $42 \text{ m s}^{-1}$  at a height of  $63 \text{ m}$  above horizontal ground, as shown in Fig. 1.1.

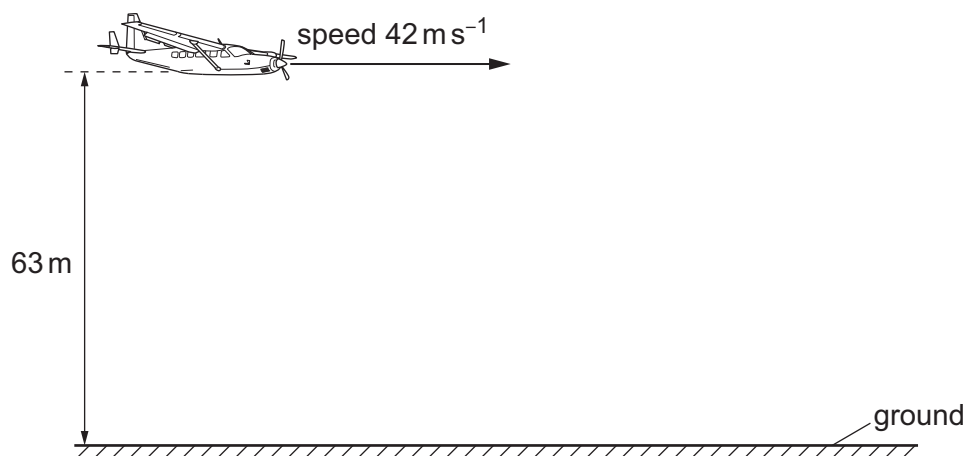


Fig. 1.1

The aircraft drops a small parcel. The parcel is released from the aircraft at the instant shown in Fig. 1.1. Air resistance is negligible.

- (i) On Fig. 1.1, draw a line to show the path of the parcel as it falls from the aircraft to the ground. [1]
- (ii) Calculate the time taken from the instant of release to the instant the parcel reaches the ground.

time = ..... s [2]





- (iii) Calculate the vertical component of the velocity of the parcel immediately before it reaches the ground.

vertical component of velocity = .....  $\text{ms}^{-1}$  [1]

- (iv) Determine the speed at which the parcel reaches the ground.

speed = .....  $\text{ms}^{-1}$  [2]

[Total: 7]

